

Kenny Mustapha

Burnaby, BC | U.S. & Canadian Citizen | 925-206-9268 | kom2@sfu.ca | [LinkedIn](#) | [GitHub](#)

EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Expected December 2027

EXPERIENCE

Student Software Engineer

May 2026 – Aug 2026

Develop for Good

Remote

- Developed a Next.js and PostgreSQL marketplace for Ikhuaya Lethu Care for the Elderly, giving community vendors an online storefront to support funding for elderly care programs.
- Translated client requirements into delivered marketplace features and implemented validation throughout vendor and storefront workflows to strengthen data integrity and reliability.

Software Engineering Intern

Jan 2026 – Apr 2026

Lillup

Remote

- Enabled audio generation from model outputs by integrating speech-generation components and implementing data conversion between model stages.
- Reduced experiment turnaround time by approximately 15% by parallelizing Python benchmarking pipelines and automating latency and audio-quality comparisons across multiple model configurations.
- Reduced manual validation effort by approximately 20% by building automated tests to compare model outputs across numerical precision settings.

Machine Learning Research Assistant

Jan 2026 – Apr 2026

SFU School of Computing Science

Burnaby, BC

- Built a Python pipeline combining YOLOv8, OpenCV camera calibration, and pinhole geometry to estimate object distance from monocular images and known object dimensions.
- Developed experimental workflows to benchmark depth estimates against physical measurements and analyze errors caused by camera parameters, object geometry, and detection bounding boxes.
- Served structured JSON experiment results through FastAPI to support reproducible experiments and downstream integration.

Software Engineering Intern

Sep 2025 – Dec 2025

Deepiri

Remote

- Reduced PostgreSQL queries by approximately 30% on cacheable API endpoints by identifying repeated queries during concurrent load testing and caching results in Redis.
- Created Python monitoring pipelines for request latency, cache hit rates, and connection pool utilization to help engineers diagnose backend bottlenecks during production incidents.
- Used profiling results to prioritize backend improvements with engineers, strengthening API reliability and performance during traffic spikes.

PROJECTS

Relay | *Go, JavaScript, PostgreSQL, Redis, Docker*

- Built a Go router that sends requests to appropriate AI models and reasoning settings to reduce token usage and cost.
- Integrated Relay into Codex through a custom plugin and built a JavaScript dashboard to track token usage, execution time, and run outcomes across seven supported models.
- Added request limits with Redis, stored usage records in PostgreSQL, and implemented automatic fallback to another provider when requests fail or time out.

PitchState | *Python, FastAPI, OpenCV, PyTorch, React, TypeScript, Jev*

- Built a soccer video analysis application that transforms uploaded clips into interactive replays with player and ball tracking, a top-down pitch view, tactical state, and next-action predictions.
- Developed an offline vision pipeline with soccer-trained models, camera calibration, motion-aware tracking, and full-clip refinement to stabilize player identities and trajectories.
- Generated player-specific action probabilities with Jev every 200 ms and synchronized the completed analysis with video playback at the source frame rate, including 60 FPS clips.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, SQL, C++

Frameworks & Libraries: FastAPI, Node.js, Express.js, Next.js, React, React Native

Databases & Developer Tools: PostgreSQL, Redis, MongoDB, AWS (EC2, S3), Prisma, Git, Docker, Codex, Claude Code

AI/ML: PyTorch, scikit-learn, OpenCV, YOLO, MediaPipe, NumPy, Pandas, pgvector

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Machine Learning, Data Mining